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Crystal Defects & Solid Solutions

Interplanar Spacing and Burgers Vector Magnitude in a Tantalum Slip System

Original statement · Problems T2 defects.pdf

9. a) Determine the interplanar spacing and the length of the Burgers vector for slipping in the slip system $(110)/[1\,\bar{1}\,1]$ *in BCC tantalum. b) Repeat assuming that the slip system is* $(111)/[1\,\bar{1}\,0]$. *Data:* $a = 3.3026$ Å. *(Solution: a)* $b = 2.860$ Å, $d_{(110)} = 2.335$ Å*; b)* $b = 4.671$ Å, $d_{(111)} = 1.907$ Å*)*

Solution: a) $b = 2.860$ Å, $d_{(110)} = 2.335$ Å; **b)** $b = 4.671$ Å, $d_{(111)} = 1.907$ Å

PHASE 1: HYPOTHESIS & PARAMETERS

Material and Lattice Parameter:

- Element: Pure tantalum (Ta), a high-density refractory metal.
- Structure: Body-Centred Cubic (BCC).
- Lattice parameter: $a = 3.3026$ Å $= 0.33026$ nm.

Slip Systems Evaluated:

1. **Real System (Part a):** Plane (110) and direction $[1\,\bar{1}\,1]$.
2. **Hypothetical System (Part b):** Plane (111) and direction $[1\,\bar{1}\,0]$.

PHASE 2: FUNDAMENTAL PHYSICS & FORMULA SELECTION

1. **Interplanar Spacing in Cubic Crystals [Session 3 Slide 53]:**

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

2. **Magnitude of the Burgers Vector ($\vec{b}$) in Cubic Lattices [Session 4 Slide 36]:**
3. Along the slip direction $[1\,\bar{1}\,1]$ (direction of maximum packing in BCC), the minimum lattice translation vector joins a vertex with the cube centre:

$$\vec{b}_a = \frac{a}{2}[1\ -1\ 1] \implies |\vec{b}_a| = \frac{a\sqrt{1^2+(-1)^2+1^2}}{2} = \frac{a\sqrt{3}}{2}$$

- Along the hypothetical slip direction $[1\,\bar{1}\,0]$ (face diagonal in BCC), the cell centre does not interrupt the diagonal; the full elementary translation vector joins two opposite vertices of a square face:

$$\vec{b}_b = a[1\ -1\ 0] \implies |\vec{b}_b| = a\sqrt{1^2+(-1)^2+0^2} = a\sqrt{2}$$

PHASE 3: ANALYTICAL DERIVATION & CALCULATION STEP-BY-STEP

1. Part a: Real System $(110)/[1\,\bar{1}\,1]$

- **Interplanar Spacing $d_{(110)}$:**

$$d_{(110)} = \frac{a}{\sqrt{1^2+1^2+0^2}} = \frac{a}{\sqrt{2}} = \frac{3.3026\ \text{Å}}{\sqrt{2}} = \frac{3.3026}{1.4142136} = \mathbf{2.3353}\ \text{Å} \approx \mathbf{2.335}\ \text{Å}$$

- **Burgers Vector Length $|\vec{b}|$:**

$$|\vec{b}| = \frac{a\sqrt{3}}{2} = \frac{3.3026\ \text{Å} \times 1.7320508}{2} = \frac{5.72027}{2} = \mathbf{2.8601}\ \text{Å} \approx \mathbf{2.860}\ \text{Å}$$

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2. Part b: Hypothetical System $(111)/[1\,\bar{1}\,0]$

- **Interplanar Spacing $d_{(111)}$:**

$$d_{(111)} = \frac{a}{\sqrt{1^2+1^2+1^2}} = \frac{a}{\sqrt{3}} = \frac{3.3026\ \text{Å}}{1.7320508} = \mathbf{1.90676}\ \text{Å} \approx \mathbf{1.907}\ \text{Å}$$

(Note that in the official solutions sheet the typographical error appears labelled as $d_{(110)} = 1.907$ Å, when it corresponds to the (111) plane).

- **Burgers Vector Length $|\vec{b}|$:**

$$|\vec{b}| = a\sqrt{2} = 3.3026\ \text{Å} \times 1.4142136 = \mathbf{4.67058}\ \text{Å} \approx \mathbf{4.671}\ \text{Å}$$

PHASE 4: PHYSICAL INTERPRETATION & VERIFICATION

Comparison of the two systems analysed:

Physical Parameter	Real System: $(110)/[1\bar{1}1]$	Hypothetical System: $(111)/[1\bar{1}0]$	Comparison
Interplanar Spacing d_{hkl}	$\mathbf{2.335}$ Å (Larger)	1.907 Å (Smaller)	$d_{(110)} > d_{(111)}$
Burgers Vector Magnitude $\lvert\vec{b}\rvert$	$\mathbf{2.860}$ Å (Smaller)	4.671 Å (Larger)	$\lvert\vec{b}_{\text{real}}\rvert < \lvert\vec{b}_{\text{hypothetical}}\rvert$
Dislocation Energy ($E \propto \lvert\vec{b}\rvert^2$**)**	$k \times (2.860)^2 = \mathbf{8.18\,k}$	$k \times (4.671)^2 = \mathbf{21.82\,k}$	**2.67 times lower energy**

Metallurgical Conclusion:

The Peierls-Nabarro stress τ_{PN} (the intrinsic resistance of the lattice to dislocation motion) decreases exponentially with the interplanar spacing d and increases with the magnitude of the Burgers vector:

$$\tau_{\text{PN}} \approx 2G \exp\left(-\frac{2\pi d}{|\vec{b}|}\right)$$

The real system $(110)/[1\bar{1}1]$ simultaneously has the **largest interplanar spacing** (2.335 Å) and the **smallest Burgers vector** (2.860 Å), making the strain energy almost three times lower and the resistance to motion minimal. For this reason, tantalum and all BCC refractory metals invariably slip in the $\{110\}\langle 111\rangle$ family.